Tongwen Wu

Contact Information

Rotman School of Management
University of Toronto
Toronto, Ontario, Canada

+1 437 667 4866
tw.wu@rotman.utoronto.ca
wtw666.github.io

Academic Appointments

University of Toronto — Nov 2024 - Present
Rotman School of Management
Postdoctoral Fellow
Supervisors: Ming Hu and Shreyas Sekar

City University of Hong Kong — Oct 2023 - Oct 2024
Postdoctoral Fellow
Supervisor: Yanzhi Li

Education

City University of Hong Kong — Sep 2019 - Oct 2023
Ph.D. in Decision Analytics and Operations
Advisor: Yanzhi Li

Sun Yat-Sen University — Aug 2015 - Jun 2019
B.Eng. in Computer Science and Technology

Research Interests

My research focuses on improving decision-making through innovative technologies and analytical methods.

- **Methodology:** Online and approximation algorithms; decision-making with predictions; data-driven optimization.
- **Applications:** Platform design; public goods allocation; order fulfillment; assortment optimization; consumer behavior modeling.

Journal Articles

[†]Authors listed alphabetically.

Ming Hu and Tongwen Wu[†] (2026)
Constant-Factor Algorithms for Revenue Management with Consecutive Stays
Operations Research. Published online.

Tongwen Wu, Zuo-Jun Max Shen, and Yanzhi Li (2026)
Anticipatory Packing
Manufacturing & Service Operations Management. Published online.

Qiaochu Fu, Woonghee Tim Huh, Yanzhi Li, and Tongwen Wu[†] (2026)
Joint Inventory and Assortment Optimization under General Choice Models
Operations Research. Forthcoming.

Working Papers

Qiaochu Fu, Woonghee Tim Huh, Yanzhi Li and Tongwen Wu[†] (2026)
Assortment and Inventory Optimization with Personalized Offering and Pricing
Under review at Operations Research.

Ming Hu, Shreyas Sekar, and Tongwen Wu[†] (2026)
Philosopher Inequality for Online Assortment Optimization
Under major revision at Operations Research.

Ming Hu, Yanzhi Li, and Tongwen Wu[†] (2025)
Diversity-Fair Online Selection
Revised and resubmitted to Operations Research.

Tongwen Wu, Woonghee Tim Huh, Menglong Li, and Yanzhi Li (2025)
A Sequential Multi-Purchase Markov Chain Choice Model: Identification, Estimation and Assortment Optimization
Under major revision at Manufacturing & Service Operations Management.

Conference Publications

Ming Hu, Shreyas Sekar, and Tongwen Wu† (2026)
Philosopher Inequality for Online Assortment Optimization
Proceedings of the 27th ACM Conference on Economics and Computation (EC 2026).

Tongwen Wu, Yu Yang, Yanzhi Li, Huiqiang Mao, Liming Li, Xiaoqing Wang, and Yuming Deng (2021)
Representation Learning for Predicting Customer Orders
ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD 2021), pp. 3735-3744.

Tongwen Wu, Zizhen Zhang, Yanzhi Li, and Jiahai Wang (2019)
Collective Mobile Sequential Recommendation: A Recommender System for Multiple Taxicabs
International Conference on Tools with Artificial Intelligence (ICTAI), pp. 1260-1264.

Tongwen Wu, Huiqiang Mao, Yanzhi Li, and Di Chen (2019)
Assortment Selection for a Frontend Warehouse: A Robust Data-Driven Approach
International Conference on Computers and Industrial Engineering (CIE 2019).

Teaching Experience

Business Statistics (CB2200) Fall 2022
Tutorial teaching, City University of Hong Kong

Operations Management (CB2201) Fall 2020 - Spring 2022
Teaching Assistant, City University of Hong Kong

Teaching interests
Operations Management, Business Statistics, Business Analytics, Revenue Management, and Data-Driven Business Analytics.

Presentations

ACM Conference on Economics and Computation, Rome	2026
University of Science and Technology of China	Dec 2025
Marketplace Innovation Workshop	2025
M&SOM Conference, Minneapolis	2024
INFORMS Annual Meeting, Seattle	2024
POMS-HK, Hong Kong, China	2023, 2022

Honors and Awards

Outstanding Academic Performance Award for Research Degree Students 2021
City University of Hong Kong

Research Tuition Scholarship 2020
City University of Hong Kong

Qualified for the International Collegiate Programming Contest (ICPC) 2018
World Finals

Gold Medal, International Collegiate Programming Contest (ICPC) 2016, 2017
Regional Contest

Gold Medal, China Collegiate Programming Contest (CCPC) 2017

Professional Service

Journal reviewer
Operations Research
ACM Transactions on Knowledge Discovery from Data

Conference reviewer
M&SOM Conference; WINE

Grant reviewer
General Research Fund, Research Grants Council of Hong Kong

Industry Experience

Alibaba Group Aug 2021 - Dec 2021
Research Intern
Digital Supply Chain Management Department

Skills and Languages

Technical: C++, Python, Java, SQL, LaTeX.
Languages: English (fluent); Chinese (native).

References

Professor Ming Hu
Postdoctoral supervisor
Rotman School of Management
University of Toronto
ming.hu@rotman.utoronto.ca

Professor Shreyas Sekar
Postdoctoral supervisor
Rotman School of Management
University of Toronto
shreyas.sekar@rotman.utoronto.ca

Professor Yanzhi Li
Ph.D. advisor
College of Business
City University of Hong Kong
yanzhili@cityu.edu.hk

Professor Woonghee Tim Huh
Research collaborator
Sauder School of Business
University of British Columbia
tim.huh@sauder.ubc.ca